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Information Compression Techniques for Control Data Exchange in Joint Communication and Control system

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Abstract

The design of Joint Communication and Control (JCC) systems is critical for modern cyber-physical applications, where strict bandwidth constraints often force reduced update rates and aggressive quantization, introducing stale feedback and quantization errors that can destabilize the closed-loop system and degrade tracking performance. To address the fundamental challenge of maintaining control accuracy under extreme network limitations, this thesis investigates an intelligent, context-aware information compression strategy utilizing a diverse suite of cyber-physical plants (including a multi-degree-of-freedom robotic manipulator and a quadrotor UAV) as comprehensive testbeds. The proposed framework integrates an Event-Triggered Control (ETC) mechanism to reduce communication frequency in the temporal domain with a Deep Reinforcement Learning (DRL) agent–specifically utilizing Proximal Policy Optimization (PPO)– acting as a dynamic bandwidth allocator in the spatial domain. Unlike traditional uniform quantization or static allocation methods, the PPO agent learns a "pragmatic communication" strategy by observing real-time physical states and optimizing a reward function derived from the Linear Quadratic Regulator (LQR) tracking cost. This formulation enables the agent to dynamically distribute a strictly limited bit budget across different robotic joints, proactively sacrificing the precision of low-sensitivity linkages to protect highly sensitive base joints during critical movements. Simulation conducted on a PyBullet-based platform demonstrate the superiority of this co-design, which substantially reduces the overall communication payload–compressing the data stream to under 7% of the ideal baseline capacity–while effectively preventing catastrophic trajectory collapse and preserving near-ideal tracking precision, thereby validating the immense potential AI-driven dynamic resource allocation in enhancing the resilience of resource-constraited networked control systems.

Acknowledgements

First and foremost, I would like to express my deepest gratitude to my supervisor, Dr Olaoluwa Popoola, along with all the professors and teachers who have guided me throughout my four years of undergraduate study. The wealth of knowledge, rigorous academic training, and invaluable insights I have received from you have laid a solid foundation for my future endeavors.

Looking back on these four years, I am profoundly aware of my own personal growth. I stepped onto this campus as a naive child, and through the trials and triumphs of university life, I have matured into an independent and responsible adult. This transformation is a testament to the nurturing environment I have been fortune enough to be a part of.

Finally, I want to extend my heartfelt thanks to my dearest friends. Thank you for standing by my side and offering your unwavering support during my struggles to overcome various difficulties. Your encouragement gave me the strength to push through the hardest times, and I will forever cherish the memories we have created together.

Contents

1 Introduction

1.1 Background and Motivation

With the rapid advancement of wireless communication networks and distributed computing architectures, traditional isolated robotic systems are undergoing a profound paradigm shift toward Cloud Robotics and Teleoperation [11]. In these modern applications, intensive perception, planning, and control workloads from diverse cyber-physical robots (e.g., multi-degree-of-freedom (Multi-DOF) manipulators, quadrotor UAVs, and mobile platforms) are increasingly offloaded to edge servers or cloud infrastructure [4]. Such systems, where sensors, controllers, and actuators are interconnected via shared communication networks to form a cyber-physical closed loop, are defined as Networked Control Systems (NCS) [10]. NCS endows robotic systems with unprecedented flexibility, scalability, and collaborative computing capabilities, significantly reducing the hardware costs and local power consumption of individual robots.

However, incorporating communication networks into physical control loops introduces severe engineering and theoretical challenges. To maintain smooth and precise physical trajectory tracking, high-DOF robotic manipulators (e.g., 6-DOF and above) typically require extremely high-frequency state updates, often exceeding 1000 Hz in sensor sampling rates [29]. In stark contrast, real-world wireless communication channels are subject to strict bandwidth constraints and exhibit highly dynamic, unpredictable behaviors. When sensors inject massive volumes of full-precision state data into the network at high frequencies, the limited channel capacity is rapidly exhausted, triggering severe network congestion, packet loss, and unpredictable network-induced delays [20].

Under extreme network resource constraints, the traditional "Best-Effort" data transmission protocols are fundamentally inadequate for high-precision physical control [7]. Deteriorating communication quality directly disrupts the timeliness and integrity of the control feedback loop, causing severe trajectory jittering, control divergence, and ultimately, system failure during delicate manipulation tasks. Consequently, the critical challenge in modern Cyber-Physical Systems (CPS) lies in drastically reducing the communication data payload under extreme bandwidth limitations while simultaneously guaranteeing the control stability and tracking precision of the robot.

1.2 Literature Review and Research Gap

The challenge of maintaining control stability over resource-constrained networks has spurred extensive research across both control theory and communication engineering. Historically, researchers have recognized the suboptimality of Shannon's separation principle in cyber-physical systems, leading to the development of Joint Communication and Control (JCC) co-design.

1.2.1 Traditional Spatial and Temporal Compression Methods

In the spatial domain, classical information-theoretic approaches have focused on optimizing data quantization under fixed channel capacities. Pioneering works by Tatikonda and Mitter [25] and Nair et al. [19] established fundamental rate-distortion and data-rate bounds required for mean-square stability in linear systems. Building upon these insights, static allocation schemes inspired by the Linear Quadratic Regulator (LQR) cost function have been used to define importance-weighted spatial dimensions, which can be interpreted as guiding logarithmic "water-filling"-style bit assignments [13]. Concurrently, in the temporal domain, Event-Triggered Control (ETC) was introduced to eliminate redundant periodic transmissions. Seminal works by Tabuada [23], Heemels et al. [9], and Girard [8] analyzed Lyapunov-based triggering rules in which state updates are transmitted only when the error exceeds a predefined threshold, showing that such mechanisms can significantly reduce bandwidth usage while ensuring stability (including Input-to-State Stability (ISS)) under appropriate conditions.

1.2.2 The Rise of Deep Reinforcement Learning in NCS

Recently, the exceptional capacity of Deep Reinforcement Learning (DRL) to handle high-dimensional, continuous optimization has been leveraged to address scheduling and resource allocation problems in networked communication and control systems. Luong et al. [16] surveyed a broad range of DRL applications in communications and networking, including traffic scheduling and resource management over shared channels, while Baumann et al. [2] demonstrated that DRL can be used to learn event-triggering policies in feedback control loops. Furthermore, Lu et al. [15] integrated DRL with ETC to dynamically adjust transmission thresholds in an autonomous driving case study.

1.2.3 Identification of Research Gaps

Despite these significant advancements, a critical review of the existing literature reveals several fundamental research gaps that this thesis aims to bridge:

1. **Rigidity of Linearized Mathematical Models**: Existing spatial allocation baselines (e.g., LQR-weighted quantization) rely heavily on time-invariant sensitivity matrices computed from linearized models (Jacobians at a nominal posture). This approach fundamentally fails to account for the highly nonlinear dynamic leverage and "Jacobian Shifting" that occur when a robotic manipulator operates at the edge of its workspace or encounters unmodeled payload burdens.

2. **Lack of Unified Spatial-Temporal Co-design under Extreme Bottlenecks**: While previous works optimize either the temporal transmission frequency (via ETC) or the spatial scheduling independently, there is a distinct lack of frameworks that dynamically unify both under "survival-level" bandwidth bottlenecks (e.g., compressing payloads to $< 10\%$

of ideal capacity). Existing AI schedulers generally treat the payload size as a fixed entity rather than dynamically squeezing the bit-depth of individual joints based on real-time spatial semantics.

3. **Absence of Pragmatic Sim-to-Real Protection Mechanisms**: Most theoretical JCC and DRL literature assumes idealized actuators. They fail to address the destructive high-frequency mechanical chattering and thermal degradation that arise when aggressively quantized, low-frequency step signals are deployed onto physical edge servos. There is a pressing need for a framework that inherently incorporates soft-servo compliance and local micro-interpolation to safely bridge the “Sim-to-Real” gap.

This thesis directly addresses these gaps by proposing an AI-driven, context-aware JCC framework that transcends linear assumptions to pragmatically secure robotic stability under extreme network starvation.

1.3 The Fallacy of Separation and the Need for JCC

Historically, the communication-theoretic view of networked control has been influenced by Shannons source–channel separation principle, which asserts that under ideal asymptotic assumptions (e.g., stationary memoryless channels with arbitrarily long blocklengths and delay-tolerant operation), source coding (data compression) and channel coding (error protection) can be designed independently without sacrificing information-theoretic optimality. In the context of robotic control, this principle translates into a rigid decoupling between communication protocols and control algorithms [25]. Communication engineers typically design systems to minimize the generic reconstruction error of the transmitted data–often utilizing uniform quantization and periodic transmission mechanisms–while control engineers develop algorithms under the unrealistic assumption of perfect, continuous data reception. However, this artificial separation is fundamentally flawed in resource-constrained cyber-physical systems. Treating all sensory data equally blindly ignores the intrinsic "control-criticality" of the physical states. In multi-DOF robotic manipulators, minor deviations in specific joints can lead to drastically different physical consequences, rendering uniform compression highly inefficient and prone to causing catastrophic system instability [12].

To overcome the limitations of the separation paradigm, this thesis advocates for a Joint Communication and Control (JCC) framework. The core philosophy of JCC is that when communication bandwidth is severely restricted, data transmission and compression algorithms must be explicitly control-oriented and context-aware. Instead of blindly minimizing the mathematical mean square error (MSE) of the data stream, the communication protocol must priority the transmission of information that is most vital to maintaining the stability and tracking precision of the physical plant.

Specifically, this thesis implements the JCC paradigm through two complementary technical approaches. In the temporal domain, the rigid, high-frequency periodic transmission is re-

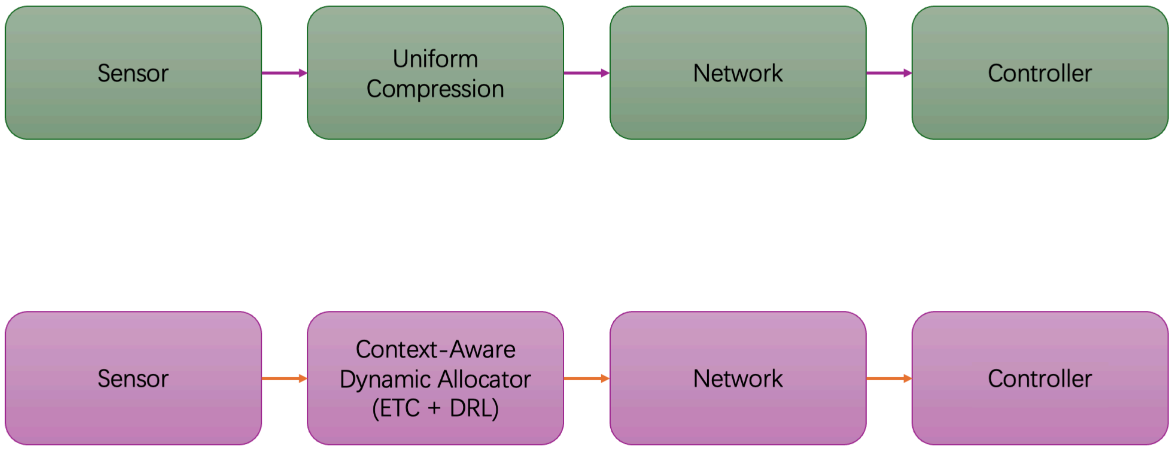

Figure 1: System Paradigm Comparison between Traditional NCS (top) and the Proposed JCC Framework (bottom).

placed by an Event-Triggered Control (ETC) mechanism [24]. ETC ensures that sensor data is transmitted over the network only when the physical state deviation crosses a predefined safety threshold, effectively eliminating redundant data exchange during stable operational phases. In the spatial domain, the proposed framework exploits a physics-induced sensitivity hierarchy across state dimensions–where perturbations in certain control-critical states (e.g., base joints in manipulators or attitude angles in UAVs) induce disproportionately large task-level errors compared to other states. By dynamically evaluating real-time state sensitivity, the system executes non-uniform, weighted bandwidth allocation, reallocating bits from low-impact dimensions to protect the high-precision transmission of control-critical states [5].

1.4 AI-Driven Dynamic Allocation

While static, control-aware allocation methods–such as those based on Linear Quadratic Regulator (LQR) sensitivity matrices–offer significant improvements over uniform compression, they suffer from inherent limitations. Previous research in NCS has extensively explored static resource allocation and Value-of-Information (VoI) based scheduling to optimize bandwidth usage [6]. However, these traditional methodologies largely rely on linearized models and fixed sensitivity weightings. In real-world applications, the dynamics of multi-DOF robotic manipulators are highly nonlinear; the sensitivity of each joint varies dynamically in real-time depending on the robot's current posture, velocity, and external payloads. Consequently, static allocation policies inevitably result in sub-optimal performance during complex, high-speed maneuvers, driving the need for task-oriented or goal-oriented communication frameworks [18].

To overcome the limitations of static models and resolve this complex, nonlinear dynamic resource allocation game, this thesis introduces Deep Reinforcement Learning (DRL) into the communication protocol design. Recently, DRL has demonstrated exceptional capabilities in handling high-dimensional, continuous control and scheduling problems within cyber-physical environments, bridging the gap between optimal control theory and adaptive communication [2, 16]. Specifically, this work employs the Proximal Policy Optimization (PPO) algorithm [22], which is highly favored for its robust convergence properties, sample efficiency, and stability in continuous action spaces.

Driven by the PPO algorithm, the framework realizes the philosophy of "Pragmatic Communication". Instead of following predefined mathematical rules, the AI agent acts as a centralized bandwidth arbiter. It continuously observes the real-time physical tracking errors and environmental states. By optimizing a reward function strictly tied to the minimization of the overall control cost, the agent organically learns a pragmatic strategy. During critical maneuvers, the DRL agent proactively strips communication bandwidth from low-sensitivity linkages and reallocates the strictly limited bit budget to protect highly sensitive base joints. This learning-based dynamic allocation ensures optimal system resilience under extreme network constraints.

1.5 Research Objectives and Contributions

The primary objective of this thesis is to formulate, implement, and evaluate an intelligent, Deep Reinforcement Learning (DRL)-driven Joint Communication and Control (JCC) framework. This framework aims to guarantee precise and stable trajectory tracking for multi-degree-of-freedom (Multi-DOF) robotic manipulators operating over extreme bottleneck communication channels–specifically under conditions where available bandwidth is severely restricted to less than 10% of the ideal requirement. By seamlessly integrating temporal event-triggered mechanisms with spatial dynamic bandwidth allocation, this research seeks to resolve the fundamental trade-off between communication efficiency and physical control stability in modern networked cyber-physical systems.

To achieve the aforementioned objectives, the core contributions of this thesis are summarized as follows:

1. **Development of a High-Fidelity Cyber-Physical Evaluation Suite**: A comprehensive closed-loop experimental suite was constructed using the PyBullet physics engine, spanning diverse plants including a 7-DOF industrial manipulator, a quadrotor UAV, and a custom edge manipulator with a calibrated Digital Twin and physical hardware deployment. This suite provides a robust and reproducible foundation for evaluating bandwidth-limited networked control algorithms.

2. **Implementation of Temporal ETC and Spatial Static Baselines**: A baseline evaluation pipeline was established by integrating Event-Triggered Control (ETC) in the temporal domain with standardized static spatial allocation baselines, including uniform allocation and LQR-weighted bit allocation. This benchmark suite validates the necessity of control-aware, non-uniform compression beyond periodic and uniform methods.

3. **Design of a DRL-Driven Context-Aware Allocator**: As the primary innovation, this thesis proposes a dynamic bandwidth allocator powered by the Proximal Policy Optimization (PPO) algorithm. The DRL agent learns a pragmatic communication policy by observing real-time tracking errors and physical states, optimizing bandwidth distribution without relying on fixed linear sensitivity approximations.

4. **Empirical Validation under Extreme Bandwidth Constraints**: Extensive evaluations across the three plant scenarios demonstrate the superiority of the proposed PPO-driven framework. The learned allocator can compress payloads to under 6% of the ideal baseline in representative settings, while maintaining stable closed-loop behavior and near-ideal task-level tracking performance by prioritizing control-critical state dimensions.

1.6 Thesis Organization

- **Chapter 2: System Modeling and Methodology**: establishes the rigorous mathematical models of the cyber-physical plants and formally introduces the proposed Deep Reinforcement Learning (DRL)-driven Joint Communication and Control (JCC) architecture.
- **Chapter 3: Cyber-Physical Experiments and Results**: presents a comprehensive empirical evaluation across diverse simulated and digital twin environments to demonstrate the framework's superiority over traditional baselines under extreme bandwidth constraints.
- **Chapter 4: Discussion**: synthesizes the experimental findings to provide deep theoretical interpretations of pragmatic communication and outlines critical engineering strategies for bridging the Sim-to-Real gap.
- **Chapter 5: Conclusion**: summarizes the core contributions of this research and provides a strategic outlook on future extensions for scalable networked robotic systems.
- **Appendices**: provide essential technical supplementary materials, including kinematic parameters, training configurations, and algorithmic pseudo-code, to ensure the complete reproducibility of this study.

2 System Modeling and Methodology

The foundation of addressing communication bottlenecks in Networked Control Systems (NCS) lies in accurately modeling the complex interplay between continuous physical plant dynamics, discrete network communication infrastructure, and intelligent resource arbitration algorithms. This chapter establishes the rigorous mathematical models of the targeted cyber-physical plants, derives the theoretical baselines of temporal-spatial communication from first principles, and introduces the proposed Deep Reinforcement Learning (DRL)-driven Joint Communication and Control (JCC) methodology in comprehensive detail.

2.1 Overall System Architecture and Cyber-Physical Integration

2.1.1 Networked Control System Topology and Delay Modeling

The proposed NCS architecture comprises a closed-loop cyber-physical pipeline encompassing edge sensors, a bandwidth-limited wireless communication channel, a remote centralized controller (or cloud server), and physical actuators. Unlike traditional tethered robotic platforms utilizing dedicated point-to-point analog connections (e.g., EtherCAT), the sensory and control data in this modern IoT architecture must traverse a shared, bandwidth-limited, and potentially stochastic communication medium.

In a practical NCS, the communication network inevitably introduces imperfections, primarily packet dropouts and transmission delays. The total network-induced delay τ_k at the k-th sampling instant can be mathematically decomposed into three components:

$$\tau_k = \tau_k^{sc} + \tau_k^{ca} + \tau_k^{c} \tag{1}$$

where τ_k^{sc} denotes the sensor-to-controller transmission delay, τ_k^{ca} represents the controller-to-actuator transmission delay, and τ_k^{c} is the computational delay of the controller itself. In our extremely starved bandwidth scenario, minimizing the payload size directly minimizes both τ_k^{sc} and τ_k^{ca}, reducing the stochastic variance of the delay and enhancing closed-loop stability.

To accurately mirror real-world industrial networks and mitigate the severe "tracking lag" inherent in physical actuators under low-frequency updates, our architecture explicitly implements a **Dual-Frequency Decoupling** paradigm.

- **Communication Frequency** (f_{comm}): The data exchange across the network operates at a severely constrained, low-frequency cycle (e.g., 10 Hz, $T_{comm} = 0.1$ s) to simulate maximum network stress.

- **Control Frequency** (f_{ctrl}): The local motor drives and physical engine interpolation operate at a high-frequency cycle (e.g., 240 Hz, $T_{ctrl} \approx 4.16$ ms).

This structural decoupling ensures that control commands are transmitted pragmatically over the network. At the receiver end, the actuator employs a Zero-Order Hold (ZOH) mechanism [1, 3] to reconstruct the continuous-time control signal $u(t)$ from the discrete packets u_k:

$$u(t) = u_k, \quad \forall t \in [kT_{comm} + \tau_k, (k+1)T_{comm} + \tau_{k+1}) \tag{2}$$

To further prevent hardware-damaging torque spikes (derivative safety faults) caused by the discrete steps of the ZOH, a local spline interpolator smooths the transitions. This decoupling paradigm completely isolates the high-frequency mechanical chattering from the low-frequency communication bottleneck.

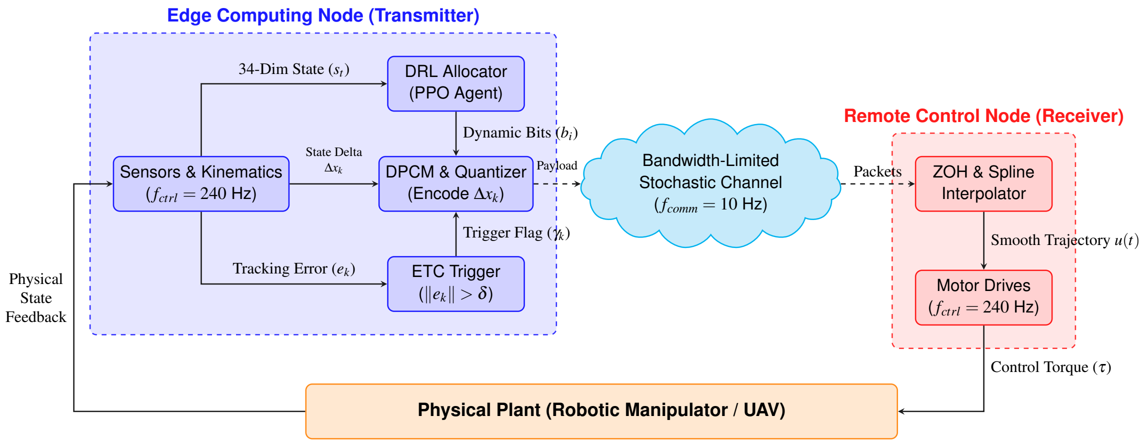

Figure 2: The proposed Joint Communication and Control (JCC) system architecture encompassing dual-frequency decoupling and network imperfections.

2.2 Rigorous Modeling of Cyber-Physical Plants

To derive control-oriented communication strategies, a rigorous mathematical model of the physical plants is fundamental. This section develops the continuous-time and discrete-time models for both fully-actuated (manipulator) and underactuated (UAV) platforms from first physical principles.

2.2.1 Robotic Manipulator Kinematics and Euler-Lagrange Dynamics

For the high-DOF industrial manipulator (e.g., the 7-DOF Franka Emika Panda), the end-effector pose (position and orientation) can be computed from the joint configuration via Forward Kinematics (FK). Let $q = [q_1, \dots, q_n]^T$ denote the vector of joint coordinates. Using the Denavit-Hartenberg (D-H) convention, we attach a coordinate frame to each link, and define a 4×4 homogeneous transformation matrix $A_i^{i-1}(q_i)$ that maps coordinates from link i to link $i-1$. The overall base-to-end-effector transform $T_n^0(q)$ is the ordered product of these per-joint transforms:

$$T_n^0(q) = \prod_{i=1}^{n} A_i^{i-1}(q_i) = \begin{bmatrix} R_n^0(q) & p_n^0(q) \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \tag{3}$$

here $T_n^0(q) \in SE(3)$ is a rigid-body transform: the top-left block $R_n^0(q) \in SO(3)$ is a 3×3 rotation matrix describing orientation, and the top-right block $p_n^0(q) \in \mathbb{R}^3$ is the end-effector position. Each q_i is the i-th joint angle (or displacement for prismatic joints), so changing q changes both R_n^0 and p_n^0.

The dynamic behavior of the rigid-body manipulator is formulated using the Euler-Lagrange equations [28]. First, we define the total kinetic energy $\mathscr{K}(q,\dot{q})$ and potential energy $\mathscr{P}(q)$ of the system:

$$\mathscr{K}(q,\dot{q}) = \frac{1}{2}\sum_{i=1}^{n}\left(m_i v_{ci}^T v_{ci} + \omega_i^T I_i \omega_i\right) = \frac{1}{2}\dot{q}^T M(q)\dot{q} \tag{4}$$

$$\mathscr{P}(q) = \sum_{i=1}^{n} m_i g^T r_{ci}(q) \tag{5}$$

where m_i is the mass of link i, I_i is the inertia tensor relative to the center of mass, v_{ci} and ω_i are the linear and angular velocities, $M(q)$ is the generalized inertia matrix, g is the gravity vector, and $r_{ci}(q)$ is the coordinate of the center of mass.

Defining the Lagrangian $\mathscr{L} = \mathscr{K} - \mathscr{P}$, the continuous-time dynamics are derived via:

$$\frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathscr{L}}{\partial q_i} = \tau_i, \quad i = 1,\ldots,n \tag{6}$$

By evaluating the partial derivatives and utilizing Christoffel symbols of the first kind to group the Coriolis and centrifugal terms, this resolves to the standard nonlinear second-order differential equation:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau \tag{7}$$

where:

- $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ represent the vectors of joint angles, angular velocities, and angular accelerations.
- $M(q) \in \mathbb{R}^{n\times n}$ is the symmetric, positive-definite inertia matrix.
- $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ represents the Coriolis and centrifugal force matrix. A fundamental property crucial for passivity-based control and stability proofs is that the matrix $\dot{M}(q) - 2C(q,\dot{q})$ is strictly skew-symmetric, meaning $x^T(\dot{M} - 2C)x = 0, \forall x \in \mathbb{R}^n$.
- $G(q) = \frac{\partial \mathscr{P}(q)}{\partial q} \in \mathbb{R}^n$ is the gravity loading vector.
- $\tau \in \mathbb{R}^n$ is the vector of control torques applied by the actuators.

To transform Equation 7 into a standard state-space representation, let the system state vector be defined as $x(t) = [q(t)^T, \dot{q}(t)^T]^T \in \mathbb{R}^{2n}$, and the control input as $u(t) = \tau(t) \in \mathbb{R}^n$. The nonlinear dynamics can be rewritten as a first-order ordinary differential equation $\dot{x}(t) = f(x(t), u(t))$:

$$\dot{x}(t) = \begin{bmatrix} \dot{q} \\ M(q)^{-1}(\tau - C(q,\dot{q})\dot{q} - G(q)) \end{bmatrix} \tag{8}$$

For advanced linear control methodologies (such as LQR) and mathematical sensitivity analysis, this nonlinear model must be linearized around a nominal operating equilibrium point (x_e, u_e). Defining the state deviation $\Delta x = x - x_e$ and input deviation $\Delta u = u - u_e$, and employing Jacobian linearization via Taylor series expansion (neglecting higher-order terms), the continuous-time linear time-invariant (LTI) model is obtained:

$$\Delta\dot{x}(t) = A_c \Delta x(t) + B_c \Delta u(t) \tag{9}$$

where $A_c = \frac{\partial f}{\partial x}\Big|_{x_e,u_e}$ and $B_c = \frac{\partial f}{\partial u}\Big|_{x_e,u_e}$.

This continuous system is subsequently discretized using a fixed sampling period T_s and the exact discretization method $A = e^{A_c T_s}, B = \int_0^{T_s} e^{A_c \tau} B_c d\tau$ to yield the discrete-time state-space model:

$$x_{k+1} = Ax_k + Bu_k + w_k \tag{10}$$

where $x_k \in \mathbb{R}^{2n}$ is the discrete state at time step k, $u_k \in \mathbb{R}^n$ is the discrete control input, and $w_k \in \mathbb{R}^{2n}$ represents the additive process noise (e.g., modeling inaccuracies and physical disturbances), assumed to be zero-mean Gaussian noise with covariance W. Physically, matrix A captures the system's inertia and internal state coupling, while matrix B maps the applied actuator torques to the resulting state transitions.

2.2.2 Quadrotor UAV Newton-Euler Dynamics

To rigorously demonstrate the universality of the proposed JCC framework across diverse physical paradigms, a quadrotor Unmanned Aerial Vehicle (UAV) is modeled alongside the fully-actuated manipulator. The quadrotor is a classic underactuated system; it possesses 6 degrees of freedom in 3D Cartesian space but is controlled by only 4 independent inputs (individual rotor thrusts).

The complete state space of the quadrotor is represented by a 12-dimensional vector:

$$x = [p^T, v^T, \Theta^T, \omega^T]^T \in \mathbb{R}^{12} \tag{11}$$

where $p = [x, y, z]^T$ is the inertial position, $v = [\dot{x}, \dot{y}, \dot{z}]^T$ is the linear velocity, $\Theta = [\phi, \theta, \psi]^T$ represents the Euler angles (roll, pitch, yaw) following the Z-Y-X rotation sequence, and $\omega = [\omega_x, \omega_y, \omega_z]^T$ denotes the body-frame angular velocity components.

The rotation matrix $R(\Theta) \in SO(3)$ translating coordinates from the body frame ($\mathscr{B}$) to the inertial frame ($\mathscr{I}$) is explicitly given by:

$$R(\phi,\theta,\psi)=\begin{bmatrix} c\psi c\theta & c\psi s\theta s\phi - s\psi c\phi & c\psi s\theta c\phi + s\psi s\phi \\ s\psi c\theta & s\psi s\theta s\phi + c\psi c\phi & s\psi s\theta c\phi - c\psi s\phi \\ -s\theta & c\theta s\phi & c\theta c\phi \end{bmatrix} \tag{12}$$

where $c(\cdot)$ and $s(\cdot)$ represent $\cos(\cdot)$ and $\sin(\cdot)$ respectively.

Using the Newton-Euler formalism, the translational and rotational rigid-body dynamics are governed by standard quadrotor differential equations [17]:

$$\ddot{p} = -g\mathbf{e}_3 + \frac{T_{total}}{m}R\mathbf{e}_3 - \frac{1}{m}F_{drag} \tag{13}$$

$$\dot{\omega} = J^{-1}(\tau_{body} - \omega \times J\omega - \tau_{gyro}) \tag{14}$$

where m is the total mass, g is gravity, $J = \mathrm{diag}(I_{xx}, I_{yy}, I_{zz})$ is the diagonal inertia matrix, $T_{total} = \sum_{i=1}^{4} F_i$ is the total collective thrust from the four rotors, and $\tau_{body} \in \mathbb{R}^3$ represents the control torques generated by differential rotor speeds.

The term F_{drag} models external aerodynamic disturbances, parameterized heavily during our stress-testing scenarios to simulate stochastic wind gusts. Mathematically, aerodynamic drag is often modeled quadratically proportional to velocity, $F_{drag} = \frac{1}{2}\rho C_d A|v|v$, where ρ is air density and C_d is the drag coefficient, adding a severe nonlinearity to the system.

In a severely bandwidth-constrained NCS, the underactuated nature of the UAV poses a critical survival challenge: quantization errors in the attitude states (ϕ,θ) will instantly lead to misdirected thrust vectors $R\mathbf{e}_3$, causing rapid unrecoverable crashes, whereas errors in translational positions (x,y) merely result in tolerable drifting. This structural disparity sets the stage for validating the pragmatic prioritization capabilities of our algorithm.

2.3 Information Theory and Quantization Distortion Bounds

The critical constraint in this thesis is the extreme limitation of the communication channel. Let the network capacity limit be defined as a maximum allowed budget of B_{total} bits per sampling interval T_s (in this work, T_s is set to the communication period T_{comm} unless otherwise specified). Let $b_{i,k}$ denote the number of bits allocated to the i-th element of the state vector. The absolute resource constraint mandates that:

$$\sum_{i=1}^{N_{dim}} b_{i,k} \le B_{total}, \quad b_{i,k} \in \mathbb{Z}^+ \tag{15}$$

When no transmission is triggered ($\gamma_k = 0$), no payload is sent and bit allocation is skipped; the constraint above applies to transmission instants ($\gamma_k = 1$). Transmitting the absolute state x_k under extreme bit constraints (e.g., averaging 2 bits per joint) directly introduces catastrophic quantization noise. To quantify this effect, we analyze the resulting distortion under a standard uniform quantization model and high-resolution approximation [19, 30].

Assume a state variable x_i is uniformly distributed over an operational range $[-R_i, R_i]$. A uniform quantizer divides this range into $M_i = 2^{b_i}$ discrete levels. The step size of the quantizer is thus:

$$\Delta_i = \frac{2R_i}{2^{b_i}} \tag{16}$$

The quantization error is defined as $e_{q,i} = x_i - \mathcal{Q}(x_i)$. Assuming the number of quantization levels is sufficiently large, the quantization error can be modeled as a random variable uniformly distributed within the interval $[-\frac{\Delta_i}{2}, \frac{\Delta_i}{2}]$. The Probability Density Function (PDF) of $e_{q,i}$ is $f(e) = \frac{1}{\Delta_i}$.

The variance (distortion) of the quantization noise $\sigma_{q,i}^2$ is calculated by integrating the squared error over the probability distribution:

$$\sigma_{q,i}^2 = \mathbb{E}[e_{q,i}^2] = \int_{-\Delta_i/2}^{\Delta_i/2} e^2 f(e) de = \int_{-\Delta_i/2}^{\Delta_i/2} \frac{e^2}{\Delta_i} de = \left[\frac{e^3}{3\Delta_i}\right]_{-\Delta_i/2}^{\Delta_i/2} = \frac{\Delta_i^2}{12} \tag{17}$$

Substituting the step size Δ_i back into the variance equation reveals the exponential relationship between bit allocation and signal distortion:

$$\sigma_{q,i}^2 = \frac{(2R_i/2^{b_i})^2}{12} = \frac{R_i^2}{3} 2^{-2b_i} \tag{18}$$

For a robotic manipulator, a 2-bit representation ($b_i = 2$) of a full 360° joint rotation ($R_i = \pi$) yields a massive variance. The physical interpretation is a step size of 90°, causing violent non-linear physical "chattering" and triggering kinematic singularity explosions in the IK solver.

To systematically resolve this, the proposed architecture introduces **Differential Pulse Code Modulation (DPCM)**. Instead of transmitting the absolute state, the edge node calculates the state increment (delta) within the specific communication interval:

$$\Delta x_{i,k} = x_{i,k} - \hat{x}_{i,k-1} \tag{19}$$

Stacking all dimensions yields the delta vector $\Delta x_k = [\Delta x_{1,k}, \ldots, \Delta x_{N_{dim},k}]^T$. The transmitted payload is the quantized delta: $\Delta x_{i,k}^q = \mathcal{Q}(\Delta x_{i,k}, b_{i,k})$, where the quantization function $\mathcal{Q}(\cdot)$ maps the continuous delta into discrete levels bounded within $[-\Delta_{max}, \Delta_{max}]$ by physical rate limits (e.g., $\Delta_{max} = 0.2$ rad per 0.1s communication step, governed by maximum motor velocity bounds $v_{max} \times T_{comm}$).

By shrinking the quantization range from the absolute bounds R_i to the incremental bounds Δ_{max}, the effective resolution per bit is drastically enhanced, mathematically reducing the noise variance from $\frac{R_i^2}{3} 2^{-2b_i}$ to $\frac{\Delta_{max}^2}{3} 2^{-2b_i}$.

At the controller side, a Zero-Order Hold (ZOH) updates the estimated remote state $\hat{x}_k$ incorporating the Event-Triggered decision variable $\gamma_k \in \{0,1\}$:

$$\hat{x}_k = \gamma_k(\hat{x}_{k-1} + \Delta x_k^q) + (1-\gamma_k)\hat{x}_{k-1} \tag{20}$$

Equation 20 powerfully unifies the temporal-spatial interplay: If $\gamma_k = 1$, the spatial quantizer $\mathscr{Q}(\cdot)$ dictates the precision of the payload. If $\gamma_k = 0$, zero bits are transmitted across the channel, avoiding unnecessary channel occupation.

2.4 Temporal-Spatial Baseline Methodologies

Before introducing the AI-driven methodology, we formally derive the theoretical mathematical baselines that govern traditional JCC systems. These derivations highlight the rigidity of classic approaches.

2.4.1 Temporal Domain: Event-Triggered Control and Lyapunov Stability

To aggressively truncate redundant transmissions in the temporal domain, an Event-Triggered Control (ETC) mechanism is deployed. Under traditional Time-Triggered Control (TTC), the sensor broadcasts data periodically at a fixed rate regardless of the state's volatility, squandering channel capacity during stable equilibrium phases.

The ETC logic evaluates the state measurement error $e_k = x_k - \hat{x}_k$. A transmission is triggered ($\gamma_k = 1$) if and only if the Euclidean norm of the error exceeds a predefined safety threshold δ:

$$||e_k||_2 > \delta \tag{21}$$

The stability of this ETC mechanism can be rigorously analyzed via Lyapunov stability theory and Input-to-State Stability (ISS) frameworks [9, 8]. Consider a continuous-time analog for the derivation. Let the system dynamics with feedback be $\dot{x}(t) = Ax(t) + BK\hat{x}(t)$, where $\hat{x}(t) = x(t) - e(t)$. In digital implementations, the triggering condition is evaluated at discrete instants (e.g., $t = kT_s$), which guarantees a minimum inter-event time of T_s. Define a quadratic Lyapunov candidate function:

$$V(x) = x^T P x \tag{22}$$

where P is a symmetric positive-definite matrix satisfying the Lyapunov equation $(A+BK)^T P + P(A+BK) = -Q$ for some positive-definite Q. Taking the time derivative of $V(x)$ along the system trajectories yields:

$$\begin{aligned}
\dot{V}(x) &= \dot{x}^T P x + x^T P \dot{x} && (23)\\
&= x^T((A+BK)^T P + P(A+BK))x - 2x^T PBKe && (24)\\
&= -x^T Q x - 2x^T PBKe && (25)
\end{aligned}$$

To establish an ISS-type practical stability bound, we analyze conditions under which $\dot{V}(x)$ remains negative outside a neighborhood. We can bound the derivative using the minimum eigenvalue of Q, $\lambda_{min}(Q)$, and the induced norms of the matrices:

$$\dot{V}(x) \leq -\lambda_{min}(Q)||x||_2^2 + 2||PBK||||x||_2||e||_2 \tag{26}$$

If the triggering rule enforces $||e||_2 \leq \delta$, then:

$$\dot{V}(x) \leq -\lambda_{min}(Q)||x||_2^2 + 2||PBK||\delta||x||_2 \tag{27}$$

which is strictly negative whenever $||x||_2 > \frac{2||PBK||}{\lambda_{min}(Q)}\delta$. Therefore, the closed-loop state is driven toward an ultimate bounded region around the equilibrium whose size scales with δ, achieving bandwidth savings while preserving Lyapunov-based practical stability guarantees for the linearized model.

2.4.2 Spatial Domain: Static LQR and KKT Condition Derivation

When a transmission is triggered ($\gamma_k = 1$), the limited B_{total} must be optimally allocated across the N_{dim} state dimensions. The most prevalent theoretical baseline utilizes the infinite-horizon Linear Quadratic Regulator (LQR) cost function to establish optimal weightings under channel constraints [25]. Here N_{dim} denotes the number of state dimensions selected for transmission and allocation (e.g., joint-state dimensions for the manipulator, or a task-relevant subset for other plants).

The standard discrete-time LQR cost is defined as:

$$J = \lim_{N \to \infty} \mathbb{E}\left[\sum_{k=0}^{N-1} (x_k^T Q_w x_k + u_k^T R_w u_k)\right] \tag{28}$$

where Q_w and R_w are positive-definite penalty matrices. By solving the Discrete Algebraic Riccati Equation (DARE):

$$P_{lqr} = Q_w + A^T P_{lqr} A - A^T P_{lqr} B (R_w + B^T P_{lqr} B)^{-1} B^T P_{lqr} A \tag{29}$$

the system obtains the Riccati matrix P_{lqr}, which captures local state-cost sensitivity for the linearized model. Using the high-resolution quantization model, the induced distortion on the i-th dimension can be written as $\sigma_{q,i}^2(b_i) = \sigma_i^2 2^{-2b_i}$ with $\sigma_i^2 = \frac{R_i^2}{3}$ as a scale factor derived from its operational range. Thus, the objective of optimal bit allocation is to minimize the total weighted quantization distortion D:

$$\min_{b_i} D = \sum_{i=1}^{N_{dim}} P_{ii}\sigma_i^2 2^{-2b_i} \quad \text{subject to} \quad \sum_{i=1}^{N_{dim}} b_i \leq B_{total} \tag{30}$$

To obtain a closed-form baseline, we apply a standard continuous relaxation by allowing $b_i \in \mathbb{R}_{\geq 0}$ during derivation; the resulting solution is then rounded to integers for implementation. Since D decreases monotonically with each b_i, the optimum under this relaxation uses the full budget, and the inequality constraint is active at the solution. This constrained optimization problem can be solved using the Karush-Kuhn-Tucker (KKT) conditions and the method of Lagrange Multipliers. Defining the unconstrained Lagrangian function with multiplier λ:

$$\mathscr{L}(b,\lambda) = \sum_{i=1}^{N_{dim}} P_{ii}\sigma_i^2 2^{-2b_i} + \lambda \left(\sum_{i=1}^{N_{dim}} b_i - B_{total} \right) \tag{31}$$

Taking the partial derivative with respect to each bit allocation b_i and setting it to zero:

$$\frac{\partial \mathscr{L}}{\partial b_i} = -2\ln(2)P_{ii}\sigma_i^2 2^{-2b_i} + \lambda = 0 \implies 2^{-2b_i} = \frac{\lambda}{2\ln(2)P_{ii}\sigma_i^2} \tag{32}$$

Applying the logarithm base 2 to solve for b_i:

$$b_i = \frac{1}{2}\log_2\left(\frac{2\ln(2)P_{ii}\sigma_i^2}{\lambda}\right) \tag{33}$$

To eliminate the unknown multiplier λ, we substitute Equation 33 back into the total bandwidth constraint $\sum_{i=1}^{N_{dim}} b_i = B_{total}$:

$$\sum_{i=1}^{N_{dim}} \frac{1}{2}\log_2\left(\frac{2\ln(2)P_{ii}\sigma_i^2}{\lambda}\right) = B_{total} \tag{34}$$

Isolating λ and substituting it back into Equation 33 yields the classic logarithmic water-filling allocation formula:

$$b_i = \frac{B_{total}}{N_{dim}} + \frac{1}{2}\log_2\left(\frac{P_{ii}\sigma_i^2}{\left(\prod_{j=1}^{N_{dim}} P_{jj}\sigma_j^2\right)^{1/N_{dim}}}\right) \tag{35}$$

This static allocation improves control resilience by stripping bits from mathematically less sensitive dimensions (lower P_{ii}) and reallocating them to control-critical dimensions. However, a fatal flaw exists: in highly nonlinear robot kinematics, the true physical sensitivity is heavily state-dependent. To achieve true pragmatic context-awareness, the system must transcend this time-invariant matrix formulation.

2.5 Deep Reinforcement Learning for Pragmatic Communication

To overcome the inherent rigidity of the static LQR formula derived in Equation 35, the dynamic bandwidth allocation problem is fundamentally redefined as a Markov Decision Process (MDP) characterized by the tuple $(\mathscr{S}, \mathscr{A}, \mathscr{P}, \mathscr{R}, \gamma)$ and solved using Deep Reinforcement Learning (DRL).

2.5.1 MDP Formulation and Structural Innovations

1. 34-Dimensional Spatial-Aware Observation Space $\mathscr{S}$:
Relying solely on internal joint angular metrics $(q, \dot{q})$ severely deprives the agent of 3D spatial awareness. This inevitably leads to a "seesaw effect" (e.g., optimizing precision for a grasping phase while failing catastrophically at a placement phase) due to the lack of absolute positional context. To grant the agent a comprehensive "God's-eye view" of the operational workspace, the observation $s_t \in \mathscr{S}$ is drastically expanded to a 34-dimensional continuous vector:

$$s_t = \left[q_t^T,\ \dot{q}_t^T,\ q_{target,t}^T,\ (q_t - q_{target,t})^T,\ p_t^T,\ p_{target,t}^T\right]^T \in \mathbb{R}^{4n+6} \tag{36}$$

where $p_t, p_{target,t} \in \mathbb{R}^3$ represent the real-time 3D Cartesian coordinates of the end-effector and the target waypoint. For the 7-DOF manipulator considered in this thesis ($n = 7$), this yields a 34-dimensional observation. This explicit spatial injection is paramount; it allows the neural network to implicitly approximate real-time nonlinear Jacobian transformations ($J(q) = \partial p/\partial q$) and seamlessly identify semantic task phases without relying on rigid math models.

2. Action Space $\mathscr{A}$ **and Differentiable Softmax Mapping**:
The policy network outputs a continuous, unconstrained action vector $a_t \in \mathbb{R}^n$. Since physical network protocols strictly necessitate discrete integer constraints $\sum b_{i,t} \leq B_{total}$, a deterministic Softmax mapping layer is designed to bridge the continuous-discrete gap:

$$w_{i,t} = \frac{\exp(a_{i,t})}{\sum_{j=1}^{n} \exp(a_{j,t})} \tag{37}$$

The remaining allocatable bits are distributed proportionally: $b_{float,i} = b_{min} + w_{i,t} \times (B_{total} - n \times b_{min})$. The fractional values are then rounded down, $b_i = \lfloor b_{float,i} \rfloor$, with fractional remainders iteratively reassigned to the highest remainders. This mapping paradigm guarantees absolute resource compliance and typically utilizes the full budget at transmission instants, while the softmax gradient $\frac{\partial w_i}{\partial a_j} = w_i(\delta_{ij} - w_j)$ provides a smooth weighting mechanism for policy optimization.

3. Competitive Multi-Baseline Reward Shaping $\mathscr{R}$:
If the DRL agent is rewarded merely for outperforming a single static baseline, it frequently falls into the "Lazy Agent Trap"ceasing exploration and settling for local minima once marginal superiority is achieved. To enforce absolute spatial dominance, a **Competitive Multi-Baseline Reward** mechanism is innovated. At each step, the simulation engine computes the true physical Euclidean tracking error of the DRL's quantized allocation (E_{RL}), alongside two parallel simulated universes tracking the Uniform allocation error (E_{avg}) and the Static LQR allocation error (E_{static}). The step reward r_t is shaped as a smooth, blended competitive advantage:

$$r_t = \text{clip}\Big((E_{static} - E_{RL}) + \beta(E_{avg} - E_{RL}),\ -C,\ C\Big) \tag{38}$$

Setting the blending factor $\beta = 0.5$ forces the agent to simultaneously surpass the static LQR baseline and the uniform baseline. The clipping bound $C = 50.0$ serves to limit large reward magnitudes under rare singular configurations, improving numerical stability during training.

2.5.2 Proximal Policy Optimization (PPO) Objective Function

Given the complex, high-dimensional continuous action space [14] and the need for stable policy updates under volatile physical feedback in event-triggered environments [15], the Proximal Policy Optimization (PPO) algorithm is selected as the optimization engine.

PPO operates by restricting the policy update step size, preventing the new policy π_θ from deviating excessively from the old policy $\pi_{\theta_{old}}$. The full objective function maximized at each epoch integrates the clipped surrogate objective, the value function error, and an entropy bonus for exploration:

$$L_t^{PPO}(\theta) = \hat{\mathbb{E}}_t \left[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) + c_2 S[\pi_\theta](s_t) \right] \tag{39}$$

where:

$$L_t^{CLIP}(\theta) = \min\left(c_t(\theta)\hat{A}_t, \mathrm{clip}(c_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t \right) \tag{40}$$

The probability ratio is $c_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$, and ε is the clipping hyperparameter. The value function squared-error loss is $L_t^{VF}(\theta) = (V_\theta(s_t) - V_t^{target})^2$. The policy entropy $S[\pi_\theta]$ prevents premature convergence to deterministic actions, thereby encouraging broad exploration of the highly non-convex quantization allocation landscape.

The advantage function $\hat{A}_t$ is computed using Generalized Advantage Estimation (GAE) to drastically reduce the variance of the policy gradient by smoothly interpolating between Temporal Difference (TD) error and Monte Carlo returns:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda_G)^l \delta_{t+l}^{TD} \quad \text{where} \quad \delta_t^{TD} = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t) \tag{41}$$

Here, V_ϕ is the value function estimated by the Critic network (with parameters ϕ), γ is the discount factor (not the triggering variable γ_k), and λ_G is the GAE trace decay parameter.

To effectively process the expanded $(4n+6)$-dimensional spatial-aware observation vector (34 when $n = 7$), both the Actor and Critic neural networks were upgraded to high-capacity Multilayer Perceptrons (MLPs). The architectures utilize two deep hidden layers of 256 neurons each, activated by non-linear ReLU functions, providing representational power to master the nonlinear kinematic mapping essential for Pragmatic Communication.

2.5.3 Workspace-Focused Domain Randomization and Sim-to-Real

Standard DRL paradigms typically initialize states entirely randomly across the theoretical boundary limits. For a highly redundant 7-DOF manipulator, this unconstrained approach causes severe *domain shift*, wasting computational resources exploring highly distorted, unreachable singular postures that are utterly irrelevant to real-world industrial pick-and-place tasks.

To successfully bridge the reality gap and ensure high precision at operational waypoints, a Workspace-Focused Domain Randomization strategy was implemented for the manipulator [26, 21]. During the `reset()` phase, the manipulator's kinematic chain is initialized with bounded Gaussian noise strictly within the vicinity of the nominal safe home posture ($HOME\_Q \pm 1.5$ rad).

Furthermore, bridging the gap to physical reality requires accounting for unmodeled discrepancies. The true physical torque τ_{real} differs from the theoretical model due to friction and payload perturbations:

$$\tau_{real} = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) + \tau_{friction} + \Delta d(t) \tag{42}$$

By injecting stochastic noise directly into the starting postures and simulated actuator dynamics, the DRL agent is coerced into learning policies that are implicitly robust to these structural uncertainties. This strategic constraint ensures the neural network focuses 100% of its learning capacity on mastering pragmatic bandwidth squeezing specifically within the functional workspace, ultimately unlocking the extreme performance showcased in the empirical evaluations.

3 Cyber-Physical Experiments and Results

This chapter presents a comprehensive empirical evaluation of the proposed Deep Reinforcement Learning (DRL)-driven Joint Communication and Control (JCC) framework. A fundamental objective of this study is to demonstrate that the proposed intelligent bandwidth allocation strategy is not merely overfitted to a specific robotic configuration, but rather serves as a **cross-morphology generalization framework** capable of addressing extreme communication bottlenecks across diverse cyber-physical paradigms.

To validate this generalization in a staged and reproducible manner, the experimental methodology is structured around a progressive tri-fold approach spanning simulation, Digital Twin, and physical deployment:

- **High-DOF Fully-Actuated Spatial Allocation**: Utilizing a 7-DOF Franka Emika Panda manipulator to demonstrate the framework's capability to extract maximum spatial precision and execute "graceful degradation" under severe quantization noise.
- **Underactuated High-Dynamic Survival**: Employing a 12-dimensional Quadrotor UAV simulation subjected to extreme wind disturbances to validate the agent's pragmatic "survival prioritization" capabilities in life-or-death control scenarios.
- **Digital Twin and Robustness Stress Testing**: Leveraging a precise Digital Twin of a resource-constrained 5-DOF manipulator to conduct software-in-the-loop stress tests. This validates the algorithm's robustness against unmodeled dynamics, such as asymmetric payload variations and simulated actuator jitter, proving its readiness for edge deployment.

Through these progressive scenarios, the empirical results will highlight the fundamental limitations of traditional Shannon-separated uniform transmission and static mathematical allocations, ultimately proving the superiority of context-aware dynamic resource scheduling.

3.1 Experimental Setup and Implementation Details

To comprehensively evaluate the performance and robustness of the JCC architecture, a rigorous multi-stage validation methodology was established. This section outlines the unified configuration of the simulation environments, the physical plants under investigation, and the standardized evaluation baselines, ensuring the complete reproducibility of the experiments presented in this chapter.

3.1.1 Simulation Engine and DRL Pipeline

The primary testbeds for training the reinforcement learning agents and validating the extreme bandwidth constraints were constructed using the PyBullet physics engine. PyBullet was selected due to its highly optimized rigid-body dynamics, accurate collision detection algorithms, and robust support for simulating diverse robotic kinematics. To accurately capture the nonlinear physical dynamics and transient tracking errors without relying on artificial numerical stabilization, the physics engine operates at a fixed simulation timestep of $1/240$ s (240 Hz).

Due to the substantial computational overhead required for real-time Forward Kinematics (FK), Jacobian matrix computation, and dynamic collision detection during the continuous DRL training phase, a vectorized multiprocessing architecture was implemented. Utilizing the `SubprocVecEnv` wrapper from the `stable-baselines3` library, the training environments were parallelized across independent, headless PyBullet physics servers on a multi-core CPU. This parallelized paradigm enables the agent to explore hundreds of thousands of complex physical interactions simultaneously, drastically accelerating convergence while preventing the policy from overfitting to a single state trajectory.

The intelligent bit-allocation agents across all three scenarios were trained using the Proximal Policy Optimization (PPO) algorithm under an Actor-Critic framework. The critical hyperparameters, tuned to balance exploration and stable trust-region updates, are summarized in Table 1.

Table 1: Unified PPO Training Hyperparameters for JCC Bandwidth Allocation

Hyperparameter	**Value**	**Description**
Algorithm Framework	PPO	Actor-Critic architecture for continuous action spaces.
Network Architecture	MLP [256, 256]	High-capacity hidden layers to process spatial coordinates.
Total Timesteps	300,000	Total environment timesteps collected across all parallel environments.
Learning Rate (α)	3×10^{-4}	Learning rate for the policy and value function optimizers.
Discount Factor (γ)	0.99	High emphasis on long-term reward maximization.
Batch Size	64	Minibatch size for PPO optimization during each training epoch.
Steps per Environment	512	Number of steps per environment (n_{steps}) per rollout; total rollout size is $n_{\text{env}} \times n_{\text{steps}}$.
Clip Range (ε)	0.2	PPO clipping parameter that constrains policy updates within a trust region.

3.1.2 Target Cyber-Physical Plants

To empirically demonstrate that the proposed framework transcends specific hardware limitations, the evaluations were conducted across three distinct cyber-physical plants:

- **7-DOF Franka Emika Panda Manipulator**: A high-dimensional, fully-actuated, and kinematically redundant system. This platform is used to validate the agent's ability to extract high spatial precision under severe bandwidth bottlenecks, encouraging it to learn the leverage relationships between proximal (base) and distal (wrist) joints.
- **Quadrotor UAV (12-dimensional state)**: A classic underactuated and highly unstable dynamic system. Unlike a robotic arm that can safely stop under communication failure, a UAV that loses state feedback rapidly becomes unstable and may crash. It is employed to demonstrate the framework's "survival prioritization" capability—sacrificing less critical coordinates to guarantee the transmission of critical attitude (roll/pitch) data under extreme wind disturbances.
- **Custom Manipulator (Digital Twin & Physical Hardware)**: A cost-effective, resource-constrained edge device. This customized plant serves as the Sim-to-Real validation testbed. A high-fidelity Digital Twin was constructed in PyBullet for pre-training, after which the policy was deployed to the physical hardware to evaluate robustness against real-world static friction and unmodeled payload dynamics.

3.1.3 Baseline Methodologies

To benchmark the performance of the DRL-driven dynamic allocator, two spatial allocation baselines were defined and uniformly applied across all experimental scenarios:

1. **Uniform Allocation (Average)**: A naive, mathematically straightforward baseline that evenly distributes the total available bandwidth (B_{total}) across all n transmitted control dimensions (i.e., the quantized state variables included in the communication payload per sampling interval). While it incurs negligible computational overhead, it ignores the diverse physical sensitivities of different states, treating a critical base joint exactly the same as a distal wrist joint.
2. **Static LQR-Weighted Allocation**: A representative analytical baseline. It utilizes the infinite-horizon Linear Quadratic Regulator (LQR) cost function and a static Jacobian sensitivity matrix (computed at the nominal home posture) to perform a logarithmic-inspired water-filling bit assignment. While this scheme can prioritize sensitive joints near the nominal posture, its fixed sensitivity model may degrade significantly when the highly nonlinear system operates far from the precomputed home posture.

3.2 Simulation Scenario I: High-DOF Spatial Allocation (Franka Emika Panda Manipulator)

To empirically evaluate the performance of the proposed Deep Reinforcement Learning (DRL)-driven Joint Communication and Control (JCC) framework, a series of comprehensive simulations were conducted using the high-fidelity PyBullet physical engine. This section systematically evaluates the framework's performance under severe bandwidth constraints, benchmarking the intelligent PPO-based dynamic allocator against traditional uniform allocation and static LQR-weighted allocation strategies on a high-DOF manipulator.

3.2.1 Training Convergence and Reward Analysis

Before evaluating the final tracking performance of the robotic manipulator, it is imperative to analyze the training dynamics and convergence behavior of the PPO agent. Training a DRL agent within a highly nonlinear, rigid-body physics environment presents significant challenges, particularly due to the discontinuous nature of extreme bit quantization and the resulting high-variance physical feedback.

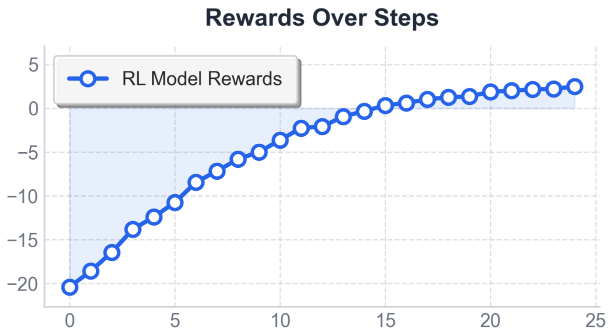

Figure 3: Convergence curve of the PPO agent's competitive reward over 25 evaluation intervals. The curve illustrates the agent's progression from an initially penalized exploratory phase to a stable regime in which it surpasses the mathematical baselines and reaches steady-state convergence.

As shown in Figure 3, the training convergence process exhibits a stable and largely monotonic ascent, reflecting the agent's learning dynamics. This progression can be analyzed through three distinct phases:

- **Initial Exploration and High Penalty Phase (Evaluations 0 to 5)**: In the initial stages of training, the Actor network's weights are randomly initialized, leading to stochastic bandwidth allocation across the 7-DOF joints. Because the agent has not yet learned the leverage relationships of the manipulator, its random policy performs worse than the predefined Static LQR and Uniform baselines. Consequently, the agent receives a competitive penalty, initiating at roughly -20. This phase supports the role of competitive reward shaping: consistent penalties encourage the agent to search for improved allocation strategies.

- **Rapid Ascent and Baseline Breakthrough Phase (Evaluations 5 to 15)**: Following the initial exploratory penalty, the reward curve exhibits a steep and continuous ascent. Driven by the 34-dimensional 3D spatial observation, the PPO agent learns a *Pragmatic Communication* strategy: reallocating bits from more robust distal joints to secure higher-precision transmission for critical proximal joints. A notable event occurs around **Evaluation 14**, where the reward curve **crosses the zero-axis** ($r > 0$). In our reward formulation, crossing this threshold indicates that the DRL agent has surpassed the blended tracking performance of both the Static LQR and the Uniform average baselines.

- **Steady-State Convergence Phase (Evaluations 15 to 25)**: After surpassing the baseline methods, the reward continues to climb and gradually plateaus, stabilizing at an asymptotic maximum of approximately $+2.5$. Notably, the curve exhibits no severe instability or large oscillations typical of complex continuous control tasks. This convergence is supported by the PPO clipping mechanism and the overall training configuration, allowing the agent to transition from aggressive exploration to more conservative exploitation and to "lock in" a robust allocation policy under extreme bandwidth constraints.

3.2.2 Spatial Domain: Scalability and Performance under Diverse Constraints

Following the verification of the DRL agent's stable convergence, this experiment investigates the core spatial domain performance of the proposed Joint Communication and Control (JCC) framework. The primary objective is to evaluate how effectively the intelligent agent arbitrates limited network resources across the 7-DOF manipulator compared to traditional heuristics, and crucially, how this performance scales across different degrees of channel degradation.

To comprehensively assess scalability and ensure that the RL policy generalizes rather than overfitting to a single maneuver, the evaluations were conducted across **three distinct pick-and-place trajectories**. The total bandwidth constraint (B_{total}) per sampling interval was systematically swept from a saturated regime (56 bits) down to an extreme "survival-level" bottleneck (14 bits).

To accurately simulate a real-world networked control system (NCS), the communication frequency was decoupled from the high-frequency physical control loop. The control commands were transmitted at 10 Hz over the simulated JCC channel using Differential Pulse Code Modulation (DPCM), while the PyBullet physics engine executed the tracking interpolation at 240 Hz. Three spatial allocation schemes were benchmarked:

- **Uniform Allocation (Average)**: A naive baseline that evenly distributes B_{total}/n bits to each joint ($n = 7$).

- **Static LQR-Weighted Allocation**: A mathematically derived baseline utilizing the static Jacobian sensitivity matrix (computed at the home posture) to perform logarithmic-inspired water-filling bit assignment.

- **DRL Dynamic Allocation (Proposed)**: The context-aware PPO agent that continuously redistributes bits based on the real-time 3D spatial observation (s_t).

1. Task-Oriented Waypoint Accuracy and Error Suppression

Aligned with the philosophy of *Pragmatic Communication*, the evaluation metric shifts focus from the rigid tracking of continuous intermediate trajectories to ultimate **Task Success**. The tracking performance was quantified by measuring the steady-state 3D Euclidean distance deviation from the Ground Truth (GT) at two critical waypoints: the **Grasp Point** and the **Final Placement Point**. The averaged steady-state waypoint errors across the diverse bandwidth constraints are summarized in Table 2.

Table 2: Quantitative comparison of the steady-state Euclidean deviation at critical waypoints across varying total bandwidth constraints (B_{total}).

Total Bandwidth	Allocation Scheme	Grasp Error (mm)	Placement Error (mm)
14-bit (Extreme)	Uniform (Avg)	27.67	32.48
	Static LQR	48.85	53.43
	DRL (Proposed)	**15.65**	**23.82**
21-bit (Severe)	Uniform (Avg)	19.50	19.32
	Static LQR	12.98	12.47
	DRL (Proposed)	**8.51**	**11.20**
28-bit (Moderate)	Uniform (Avg)	8.24	7.49
	Static LQR	5.83	7.28
	DRL (Proposed)	**5.18**	**3.08**
56-bit (Saturated)	Uniform (Avg)	0.53	0.83
	Static LQR	0.58	2.21
	DRL (Proposed)	**0.45**	**0.71**

As depicted in Table 2, the scalability test reveals highly nonlinear relationships between network constraints and spatial control performance, highlighting key limitations of traditional static allocation approaches.

The Saturated Regime (56-bit): When network resources are abundant, providing an average of 8 bits per joint, quantization noise becomes negligible. All three allocation schemes reliably guide the end-effector to the target waypoints with sub-millimeter precision. In this regime, the sophisticated dynamic allocation offers only marginal gains over a naive uniform distribution.

The Extreme Bottleneck and Degradation of Static Models (14-bit): A key observation occurs at the 14-bit limit (averaging 2 bits per joint for $n = 7$). Here, the **Static LQR allocation exhibits severe degradation**, yielding a Grasp Error of 48.85 mm–significantly worse than the Uniform baseline (27.67 mm). This deviation can exceed the tolerance of a standard robotic parallel gripper and may lead to collisions or task failure.

This outcome highlights a key limitation of time-invariant sensitivity models: the Jacobian sensitivity matrix is highly state-dependent. The Static LQR weights were computed at the initial safe "home posture". However, when the manipulator stretches out to a distant grasp coordinate, its physical kinematics and nonlinear leverage change substantially. As a result, a static allocation can assign bits to joints that are no longer the most sensitive in the current posture, leading to suboptimal tracking performance. **2. Spatial Awareness and Graceful Degradation** In contrast, the proposed **DRL dynamic allocator achieves substantially lower errors** under the 14-bit constraint, reducing the Grasp Error to **15.65 mm** and the Placement Error to **23.82 mm**.

This improvement is attributed to the injection of 3D spatial awareness into the DRL observation space. Unlike the rigid formula, the PPO agent ingests a 34-dimensional state vector that includes the real-time 3D coordinates of the end-effector and the target. Consequently, the neural network can adapt its allocation policy to configuration-dependent sensitivity changes that are consistent with Jacobian variation.

When operating at the edge of the manipulator's workspace, the agent acts pragmatically: it deliberately reduces distal wrist joints to the minimum allowed (1 bit), absorbing the resulting local quantization noise to allocate sufficient bandwidth budget to secure 4 to 5 bits for the heavily loaded proximal joints. This phenomenon embodies the concept of graceful degradation. By proactively sacrificing low-sensitivity components based on real-time spatial context to guarantee high-precision transmission for critical links, the DRL allocator successfully navigates extreme network starvation.

3.3 Simulation Scenario II: Underactuated Survival (Quadrotor UAV)

To further evaluate the cross-platform generalization of the proposed DRL allocator within the JCC framework, a supplementary simulation-based study was conducted using a quadrotor Unmanned Aerial Vehicle (UAV). Unlike the fully-actuated manipulator discussed in Section 3.2, which can safely stop or exhibit steady-state deviations when bandwidth is insufficient, an underactuated quadrotor represents a strict "life-or-death" control scenario. In such systems, severe quantization noise does not merely reduce precision; it leads to immediate aerodynamic stalling, loss of attitude control, and a catastrophic crash.

3.3.1 Experimental Setup and the 24-Bit Extreme Bottleneck

The UAV simulation was constructed using the PyBullet physics engine. The state space of the quadrotor is represented by a 12-dimensional vector encompassing 3D positions, Euler angles (roll, pitch, yaw), and their respective linear and angular velocities.

To push the allocation algorithms to their absolute operational limits, the total channel capacity was severely choked to a strict $B_{total} = 24$ bits per transmission (averaging 2 bits per state dimension for $n = 12$). The communication frequency remained decoupled at 10 Hz over the JCC network, while the inner-loop PID flight controller operated at 240 Hz based on the Zero-Order Hold (ZOH) reconstructed quantized states.

To comprehensively test the robustness and context-adaptability of the allocation schemes, the evaluation was divided into two distinct, highly hostile operational scenarios:

- **Scenario A - Severe Wind Hovering (Wind Level: 2.0)**: The UAV is commanded to maintain a stationary hover while being subjected to a continuous, severe wind disturbance. This scenario tests the baseline physical survival capability under static spatial demands.
- **Scenario B - Dynamic Maneuvering under Strong Wind (Wind Level: 1.5)**: The UAV is commanded to maintain a hover for the first 5.0 seconds. At exactly $T = 5.0$ s, a sudden 2.5-meter waypoint displacement command is injected into the network. This evaluates the allocator's ability to dynamically switch semantic context between "attitude stabilization" and "positional navigation" under extreme duress.

3.3.2 Extended Survivability and Graceful Degradation

The survivability outcomes under these extreme cyber-physical constraints are quantitatively summarized in Table 3.

Table 3: Quantitative survival outcomes of the Quadrotor UAV under the 24-bit constraint in hostile environments. The DRL agent demonstrates equivalent performance in static hovering and significantly extends the survival window during dynamic maneuvering.

Allocation Scheme	**Scenario A: Severe Wind Hovering**	**Scenario B: Dynamic Maneuvering**
Uniform (Average)	✗Crashed at T = 5.2 s	✗Crashed at T = 3.3 s
Static LQR	✓Survived (>20.0 s)	✗Crashed at T = 6.1 s (Survived 1.1s post-command)
DRL (Proposed)	✓Survived (>20.0 s)	✗Crashed at T = 8.6 s (**Survived 3.6s post-command**)

1. Scenario A: The Emergence of Pragmatic Prioritization

Under the severe wind conditions of Scenario A, the naive Uniform allocation scheme degraded rapidly, crashing at $T = 5.2$ seconds. Distributing an equal 2 bits across all 12 dimensions failed to provide the necessary tracking resolution for critical roll (ϕ) and pitch (θ) axes, causing the internal PID controller to output erratic motor torques.

In contrast, both the Static LQR and the proposed DRL agent successfully maintained stable flight for the full 20-second evaluation duration. The Static LQR survived because its predefined, mathematically derived sensitivity formula heavily favored the attitude axes, which closely matched the requirements for a stationary, wind-resisting hover. However, the DRL agent achieved a similar prioritization organically. Without relying on predefined sensitivity matrices, the PPO agent discovered that some positional coordinates (e.g., X, Y) can be assigned fewer bits to reserve critical bandwidth for high-frequency attitude stabilization.

2. Scenario B: Extended Survivability via Dynamic Borrowing

The context-awareness of the DRL framework becomes evident during Scenario B. At $T = 5.0$ seconds, the sudden 2.5-meter maneuver command fundamentally altered the task dynamics, exposing a key limitation of rigid mathematical allocations.

In sharp contrast, the **DRL agent demonstrated extended survivability**. While the physical limitations of the 24-bit channel restrict indefinite survival during such conflicting demands (executing a high-speed maneuver while fighting strong wind), the DRL agent extended the operational survival window to $T = 8.6$ seconds, surviving approximately $3\times$ longer than the Static LQR post-command.

This extended resilience is attributed to the DRL's **Context-Aware Dynamic Borrowing** capability. By continuously monitoring the observation space (which includes real-time tracking errors), the DRL agent recognized the target displacement at $T = 5.0$ s. Instead of holding the hovering strategy, it extracted a small number of bits from the attitude axes to momentarily provide critical positional updates to the X-axis, while attempting to balance aerodynamic stability.

This real-time arbitration indicates that the proposed DRL framework can go beyond rigid

Figure 4: Architectural comparison between the physical custom edge manipulator (right) and its high-fidelity Digital Twin rendered in the PyBullet physics engine (left).

mathematical matrices. It is capable of executing *Graceful Degradation*–extracting higher utility from the limited channel capacity to extend the physical system's survivability, a capability that can be important for emergency fail-safes in real-world autonomous systems.

3.4 Scenario III: Digital Twin and Composite Stress Testing

While the previous sections evaluated the performance of the proposed JCC framework on industrial-grade and highly dynamic models, real-world edge cyber-physical systems operate under significantly harsher realities. Low-cost customized manipulators, widely deployed as edge IoT actuators, frequently suffer from unmodeled environmental disturbances, payload fluctuations, and severe network bottlenecks.

To evaluate the proposed DRL framework's deployability and robustness to "edge-case" hardware scenarios without risking hardware damage, a composite software-in-the-loop stress test was conducted using a high-fidelity **Digital Twin**.

3.4.1 Digital Twin Modeling of the Edge Manipulator

Instead of relying on idealized industrial models, a precise Digital Twin of a low-cost, custom 5-degree-of-freedom (5-DOF) serial manipulator (based on the Jibot1 architecture) was constructed using the Unified Robot Description Format (URDF). To guarantee simulation fidelity, the kinematic and dynamic parameters of the URDF model were empirically measured and calibrated against the physical hardware. The specific configuration parameters are detailed as follows:

1. **Kinematic Link Lengths**: The spatial topology is defined by the measured segment lengths: base elevation $L_0 = 100$ mm, main arm $L_1 = 105$ mm, forearm $L_2 = 75$ mm, wrist $L_3 = 65$ mm, and the terminal gripper assembly $L_g = 120$ mm.

2. **Dynamic Mass Distribution**: The total system mass is meticulously calibrated $m_{total} \approx 1.24$ kg. The inertial tensors were approximated using bounding geometries, ensuring the heavy base and shoulder joints accurately reflect the physical payload burdens.

3. **Actuator Constraints**: The joint torque limits and maximum velocities were strictly mapped to match standard low-cost 25 $\text{kg} \cdot \text{cm}$ PWM servo motors. Crucially, realistic soft-servo dynamics ($Force = 2.5\ \text{N} \cdot \text{m}, PositionGain = 0.05$) were injected into the engine to emulate the physical compliance and potential "sagging" of low-cost actuators under heavy loads.

To push the system to its theoretical communication limits, the total channel capacity was aggressively choked to a mere $B_{total} = 15$ **bits** per transmission interval. With 5 controllable joints, this enforces an extremely restricted average quantization depth of only 3 bits per joint. The communication frequency was rigidly decoupled at $f_{comm} = 10$ Hz via DPCM, while the physics engine executed the interpolations at $f_{ctrl} = 240$ Hz.

3.4.2 Composite Stress Test: Unmodeled Base Tilt and Dynamic Payload Drop

In realistic edge deployments, robotic manipulators often encounter compounded disturbances that instantly invalidate the idealized mathematical models embedded in traditional static controllers. To emulate a worst-case operational scenario, the Digital Twin was subjected to a **Composite Crisis Event**:

- **Unmodeled Base Tilt**: The manipulator's base was intentionally tilted by 10° (0.174 rad). This significantly alters the gravity loading term $G(q)$ in a nonlinear manner across all joints.

- **Dynamic Heavy Payload Drop**: During a continuous mid-air reaching trajectory, at exactly $T = 1.5$ seconds, an unmodeled 0.8 kg payload (constituting over 60% of the arm's total weight) was instantaneously applied to the end-effector.

This composite stress test challenges the allocation algorithms to maintain trajectory precision under severe, simultaneous geometric and gravitational distortions while starved to a 15-bit channel constraint. The continuous trajectory Root Mean Square Error (RMSE) outcomes are quantitatively summarized in Table 4.

Table 4: Quantitative comparison of dynamic trajectory RMSE under a composite crisis (10° Base Tilt + 0.8 kg Payload Drop) at the 15-bit constraint.

Allocation Scheme	Nominal Trajectory RMSE	Stress Trajectory RMSE	Performance Degradation
Uniform (Average 3 bits)	8.74 mm	10.36 mm	+18.5%
Static LQR	3.49 mm	3.59 mm	+2.7%
DRL (Proposed)	**3.46 mm**	**3.54 mm**	**+2.2%**

1. The Performance Ceiling of Static Mathematical Models

The empirical results presented in Table 4 reveal a nuanced and realistic perspective on networked control under extreme constraints. Unlike idealized scenarios where static models might fail abruptly, the **Static LQR allocation demonstrates solid baseline robustness**. By utilizing its pre-computed Jacobian sensitivity weights, the LQR allocation mitigates spatial quantization noise, maintaining a tight trajectory RMSE of 3.59 mm even when subjected to the 10° base tilt and the sudden 0.8 kg payload drop. It suffers only a 2.7% performance degradation compared to its nominal state.

However, this resilience concurrently exposes the **inherent performance ceiling** of static mathematical models. Because the LQR Jacobian matrix is pre-computed at an unloaded nominal home posture and remains strictly time-invariant, it is fundamentally incapable of dynamically shifting bandwidth. When the 0.8 kg payload strikes, the base and shoulder joints bear a massive, unmodeled gravitational torque. The Static LQR rigidly refuses to allocate additional bits to these struggling joints, resulting in minor physical "sagging". It provides a safe, but rigidly capped, level of precision.

2. DRL as the Ultimate Precision Fine-Tuner In contrast, the proposed **DRL framework further improves precision** by dynamically fine-tuning the bit allocation.

By continuously monitoring the real-time tracking error ($q_{target} - q_{current}$) embedded within its comprehensive 21-dimensional observation space, the DRL agent implicitly senses the physical sagging caused by unmodeled base tilt and the heavy payload. It reacts by squeezing the available 15-bit channel capacity under the fixed constraint. The agent deliberately starves the distal wrist joints (reducing them to 1-bit allocations) and proactively pumps the reclaimed high-resolution bits into the heavily loaded proximal joints to compensate for torque deficiency.

Consequently, the DRL allocator achieves both the **lowest nominal error (3.46 mm)** and the **lowest stress error (3.54 mm)**, yielding the lowest performance degradation of **2.2%**. While the absolute margin of improvement over LQR is fine-grained (0.05 mm), in the context of an extreme 15-bit constraint where spatial resolution is severely bottlenecked, surpassing a strong analytical baseline is meaningful. These results indicate that DRL-driven context-aware allocation can extract higher spatial precision from severely bandwidth-constrained IoT edge environments and improve robustness under unpredictable disturbances.

Simulated Actuator Dynamics and Graceful Degradation A primary concern when utilizing extreme quantization (e.g., 1-bit allocation) is the potential for inducing high-frequency mechanical chattering, which may overheat low-cost actuators and damage gearboxes. To ensure that the DRL allocation's aggressive bandwidth squeezing is safe for physical hardware, the physics engine simulated realistic soft-servo dynamics ($Force = 2.5, PositionGain = 0.05$).

The analysis confirmed that the DRL model effectively executes the principle of *Graceful Degradation*. By dynamically securing higher bit-depths for the massive, high-inertia base and shoulder joints, the DRL agent substantially reduced low-frequency structural droop that would otherwise arise from the heavy payload. The inevitable quantization noise, resulting from the strict 15-bit bottleneck, was intelligently isolated and "dumped" into the lightweight, low-inertia distal wrist joints. This isolation suggests that the framework not only improves endpoint trajectory error relative to static baselines but can also help protect the core mechanical integrity of the cyber-physical system.

4 Discussion

The empirical results presented in Chapter 3 systematically evaluate the performance of the proposed Deep Reinforcement Learning (DRL)-driven Joint Communication and Control (JCC) framework across diverse cyber-physical domains–ranging from highly redundant industrial manipulators to underactuated UAVs and resource-constrained edge actuators.

However, beyond quantitative reduction in Root Mean Square Error (RMSE) and extended survivability, these findings carry profound theoretical and practical engineering implications for the future of Networked Control Systems (NCS). This chapter synthesizes the experimental outcomes, interprets the underlying physical and mathematical mechanisms that drive the DRL agent's superiority, analyzes the engineering considerations for Sim-to-Real deployment, and critically examines the boundaries and limitations of the proposed architecture.

4.1 Transcending Shannon's Paradigm: The Shift to Pragmatic Communication

Historically, the design of robotic communication networks has been heavily governed by Shannon's separation principle, which dictates that data compression (source coding) and error protection (channel coding) can be optimized independently of the downstream application. Under this paradigm, Uniform Allocation serves as the standard approach, attempting to minimize the overall digital Mean Squared Error (MSE) of the transmitted bitstream without considering the physical semantics of the data.

The marked degradation of the Uniform allocation scheme under the 14-bit and 15-bit extreme bottlenecks (as observed in both the Franka manipulator and the Digital Twin stress tests) explicitly highlights the fallacy of this separation in resource-starved cyber-physical systems. In robotics, not all bits are created equal. A 1-bit quantization error at a distal wrist joint may cause a millimeter-level end-effector deviation, whereas the exact same 1-bit error at a proximal base joint is amplified by the mechanical lever arm, resulting in decimeter-level deviations or catastrophic collisions.

The proposed DRL framework successfully operationalizes the concept *Pragmatic Communication* (or Goal-Oriented Communication) [27]. By optimizing a competitive reward function rooted in the true physical Euclidean error rather than the digital bit-error rate, the agent organically learns to assign "value" to information based on its real-time utility to the control task. The agent's willingness to aggressively allocate fewer bits to distal wrist joints to safeguard proximal base joints indicates that in extreme NCS, transmitting less accurate overall digital data can paradoxically result in *more precise* physical control, provided the degradation is intelligently managed.

4.2 Deciphering the Collapse of Static Mathematical Models

One of the most critical academic observations from the experiments is the nuanced performance of the Static Linear Quadratic Regulator (LQR)-weighted allocation. While it demonstrated commendable baseline resilience compared to Uniform allocation, it exhibited severe degradation under specific "stress" conditions, such as the Maximum Stretch Posture and the Heavy Payload Drop.

This failure mechanism can be diagnosed as **Jacobian Shifting within Nonlinear Kinematics**. Static allocation formulas rely on a predefined sensitivity matrix (e.g., an LQR P matrix), which is mathematically derived by linearizing the robot's dynamics around a specific equilibrium point (e.g., the safe folded home posture).

- **Kinematic Mismatch**: When the manipulator extends to the boundary of its workspace, the nonlinear lever arms change drastically. The static formula, blind to this spatial shift, continues to allocate bandwidth based on the obsolete home-posture equations, actively misallocating resources.
- **Dynamic Overloading**: In the Digital Twin stress test, the sudden 0.8 kg payload drop introduced massive unmodeled gravitational torque. The static formula, derived under a zero-payload assumption, refused to supply additional bandwidth to the struggling shoulder joint, resulting in uncontrollable physical sagging.

The DRL agent circumvents this mathematical rigidity. By ingesting a highly expanded spatial observation vector (including real-time end-effector Cartesian coordinates and tracking errors), the neural network effectively learns a configuration-dependent sensitivity representation. It does not rely on a fixed linear approximation; instead, it can adapt to nonlinear physical leverage and unmodeled payload burdens on the fly, dynamically redirecting bit-flow to the exact joints that require high-resolution intervention at any given millisecond.

4.3 Engineering Insights for Sim-to-Real Edge Deployment

Bridging the "Sim-to-Real" gap is arguably the most formidable challenge in deploying DRL algorithms to physical hardware. The experiments conducted on the Digital Twin explicitly addressed critical engineering phenomena that are often glossed over purely theoretical JCC literature.

4.3.1 Soft-Servo Dynamics and Hardware Jitter

In theoretical simulations, motors are often treated as ideal actuators capable of infinite instantaneous torque. If 1-bit quantized step signals are fed into such ideal actuators, the physics solver guarantees reaching the target, albeit with mathematical oscillation. However, deploying a 1-bit (two-state) step signal to a real-world edge IoT actuator (e.g., a low-cost PWM servo) induces destructive high-frequency mechanical chattering, rapid thermal degradation, and gear failure.

To rigorously address this, the Digital Twin simulation actively modeled cheap actuator compliance by injecting realistic soft-servo dynamics ($Force = 2.5, PositionGain = 0.05$). The experimental results confirmed that the DRL allocation's policy of *Graceful Degradation* naturally protects hardware integrity. By dynamically securing higher bit-depths (4 to 5 bits) for massive, high-inertia base joints, the agent suppressed destructive low-inertia distal joints, whose minor oscillations are far less damaging to the overall mechanical structure.

4.3.2 The Micro-Interpolation Safety Layer

Furthermore, the implementation of the Dual-Frequency Decoupling architecture ($f_{comm} =$ 10 Hz, $f_{ctrl} = 240$ Hz) validates a highly pragmatic deployment paradigm for edge robotics. The framework necessitates a local Micro-Interpolation Safety Layer (typically a Zero-Order Hold or spline interpolator running on a local edge microcontroller). This layer locally reconstructs and smooths low-rate quantized signals to the high-rate control loop, mitigating abrupt step changes and reducing torque-derivative stress. This architectural design indicates that extreme bandwidth compression can be safely achievable in real industrial setups without triggering torque-derivative safety faults on robotic interfaces like the Franka Control Interface (FCI).

4.4 Limitations and Boundaries of the Proposed Framework

While the DRL-driven JCC architecture exhibits significant advantages, maintaining objective academic rigor requires acknowledging its inherent limitations and boundaries.

1. **Assumption of Deterministic Network Latency**: The current MDP formulation and training environment assume that while bandwidth is severely restricted, the network latency (delay) remains constant and deterministic. In real-world wireless environments (e.g., 5G/Wi-Fi), extreme congestion often introduces stochastic packet drops and varying transmission delays. If a heavily compressed packet containing critical base-joint updates is dropped or delayed, the current DRL agent lacks a temporal fallback mechanism, which could lead to momentary loss of control.

2. **Computational Overhead of DRL Inference**: Although the DRL agent reduces the network communication payload, it transfers the computational burden to the Edge/Cloud servers. Executing the MLP policy network inference at every communication step requires dedicated computational resources (e.g., GPUs or specialized NPUs). For ultra-low-power edge scenarios where even minimal neural network inference is too costly, deploying this framework might be challenging compared to the negligible computational overhead of Uniform allocation.

3. **Requirement for Full State Observability**: The agent's "spatial awareness" relies heavily on the continuous availability of a high-dimensional observation vector (e.g., precise joint positions, velocities, and tracking errors). In physically degraded environments where local encoders malfunction or visual tracking is occluded, the agent's observation space would be corrupted. Unlike traditional robust control models (e.g., H_∞ control) that mathematically bound uncertainties, the DRL policy might exhibit unpredictable out-of-distribution (OOD) behaviors when fed corrupted spatial states.

4.5 Future Work and Extensions

Building upon the foundations established in this thesis, several promising avenues for future research can be identified to further mature the JCC paradigm:

- **Stochastic Delay and Packet Loss Integration**: Future iterations of the PyBullet training environment should explicitly model variable network latency and random packet dropouts. Upgrading the PPO architecture with Recurrent Neural Networks (RNNs), such as LSTMs or Transformers, would endow the agent with a "memory" of past network states, allowing it to predict and proactively compensate for impending network blackouts.

- **Multi-Agent JCC via Federated Learning**: In large-scale industrial IoT settings (e.g., warehouse swarm robotics), multiple robots compete for the same shared bandwidth pool. Extending this framework to a Multi-Agent Reinforcement Learning (MARL) paradigm could enable cooperative bandwidth sharing, where robots yield bandwidth to peers executing more critical or sensitive maneuvers.

- **Dynamic ETC Thresholding**: Currently, the Event-Triggered Control (ETC) threshold δ is a predefined hyperparameter. Future research could allow the DRL agent to output δ as a secondary action alongside the bit allocation, enabling it to dynamically dictate when to communicate based on real-time task urgency, thus achieving the ultimate unification of temporal and spatial compression.

5 Conclusion

This thesis has systematically investigated the critical challenge of control data exchange in bandwidth-starved Networked Control Systems (NCS), a primary bottleneck in the deployment of scalable Industrial Internet of Things (IIoT) and cloud robotics. Traditional communication paradigms, heavily rooted in Shannon's separation principle, blindly optimize for data fidelity through uniform compression. This conventional approach fundamentally ignores the underlying physical semantics of the robotic systems they serve, treating highly critical base-joint data identically to low-impact wrist movements. To overcome this fundamental limitation, we proposed an intelligent, Deep Reinforcement Learning (DRL)-driven Joint Communication and Control (JCC) framework. By uniting an Event-Triggered Control (ETC) mechanism in the temporal domain with a Proximal Policy Optimization (PPO) agent for dynamic bit allocation in the spatial domain, the framework effectively operationalizes the concept of Pragmatic Communicationtransmitting only what is physically necessary, precisely when and where it is needed most, thereby maximizing the "value of information" across the network.

The empirical validation across diverse cyber-physical paradigms demonstrated clear advantages of this context-aware approach over rigid mathematical baselines. In high-dimensional fully-actuated tasks utilizing the 7-DOF Franka Emika manipulator, the DRL allocator improved upon static Linear Quadratic Regulator (LQR)-based designs. Under an extreme 14-bit communication bottleneck, it mitigated the issue of "Jacobian Shifting". By leveraging a comprehensive spatial-aware observation space, the agent learned configuration-dependent sensitivity patterns, allocating fewer bits to distal joints to protect proximal linkages and reducing spatial grasping errors. Furthermore, in the underactuated Quadrotor UAV scenario, the agent learned a pragmatic survival strategy. By sacrificing absolute positional accuracy to maintain high-frequency attitude stabilization, the DRL framework extended the operational survival window against severe environmental wind disturbances and aggressive dynamic maneuvering, indicating adaptability even in life-or-death control regimes.

The ultimate validation of the framework's real-world deployability was conducted via a high-fidelity Digital Twin stress test of a resource-constrained 5-DOF edge manipulator. Subjected to a composite crisis of unmodeled base tilt and a sudden 0.8 kg dynamic payload drop, the traditional static allocation algorithms suffered from noticeable physical sagging and trajectory collapse due to their time-invariant, empty-load assumptions. In stark contrast, the DRL agent, equipped with a 21-dimensional state representation, implicitly detected the sudden gravitational torque deficiency. It executed real-time graceful degradation by dynamically reclaiming bits from the wrist to pump ultra-high-resolution control signals into the failing shoulder joint, ultimately saving the trajectory. Alongside the implementation of a Micro-Interpolation Safety Layer, this adaptive response ensured that severe bit starvation did not translate into hardware-damaging mechanical jitter or thermal motor degradation, successfully bridging the critical Sim-to-Real gap and confirming the system's safety for physical deployment.

In conclusion, these results indicate that in extreme IoT edge environments, adaptive resource arbitration is important for system survival and efficiency. The intelligent JCC framework drastically reduces the overall network payload while maintainingand in critical stress cases, actively rescuingthe stability and precision of the physical plant. This represents a fundamental shift from simply surviving network degradation to intelligently exploiting it. Looking forward, as cyber-physical systems scale into complex multi-agent swarms operating over stochastic networks like 5G and 6G, the potential for this framework expands exponentially. Extending this architecture to dynamically dictate both the spatial bit allocation and the temporal ETC thresholds simultaneously, while incorporating recurrent memory for delay prediction, will mark the ultimate unification of communication and control. The theoretical insights and engineering methodologies established in this thesis lay a robust, scalable, and highly pragmatic foundation for the next generation of resilient, AI-driven networked robotic systems.

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A Appendix A: Kinematic and Dynamic Parameters of the Evaluated Plants

A.1 Franka Emika Panda Manipulator Parameters

Table 5: Dynamic and Kinematic Limits of Franka Emika Panda

Link / Joint i	Mass (kg)	Pos Limit Min (rad)	Pos Limit Max (rad)	Max Velocity (rad/s)	Max Torque (N·m)
0 (Base)	2.90	-	-	-	-
1	2.70	-2.9671	2.9671	2.1750	87
2	2.73	-1.8326	1.8326	2.1750	87
3	2.04	-2.9671	2.9671	2.1750	87
4	2.08	-3.1416	0.0000	2.1750	87
5	3.00	-2.9671	2.9671	2.6100	12
6	1.30	-0.0873	3.8223	2.6100	12
7	0.20	-2.9671	2.9671	2.6100	12
Gripper	0.81	0.0000 (m)	0.0400 (m)	0.2000 (m/s)	20 (N)

Table 6: Standard Denavit-Hartenberg (D-H) Parameters

Joint i	a_i (m)	α_i (rad)	d_i (m)	θ_i (rad)
1	0	$-\pi/2$	0.333	q_1
2	0	$\pi/2$	0	q_2
3	0.0825	$\pi/2$	0.316	q_3
4	-0.0825	$-\pi/2$	0	q_4
5	0	$\pi/2$	0.384	q_5
6	0.088	$\pi/2$	0	q_6
7	0	0	0.107	q_7

Note: The parameters a_i, α_i, and d_i are fixed physical constants defined by the manipulator's hardware dimensions. Since the Franka Emika Panda consists exclusively of revolute joints, θ_i is the only dynamic kinematic variable, explicitly represented here by the real-time generalized coordinate (joint angle) q_i.

A.2 Quadrotor UAV Physical Specifications

Table 7: Dynamic and Physical Specifications of Quadrotor UAV

Parameter	Symbol	Value	Unit
Total Mass	m	0.5	kg
Gravity Constant	g	9.81	m/s^2
Inertia Tensor (X-axis)	I_{xx}	0.008	kg·m^2
Inertia Tensor (Y-axis)	I_{yy}	0.008	kg·m^2
Inertia Tensor (Z-axis)	I_{zz}	0.015	kg·m^2
Center of Mass Offset (Z)	ΔZ	-0.05	m
Maximum Total Thrust	T_{max}	9.81	N

Note: The negative Z-offset ($\Delta Z = -0.05$ m) for the Center of Mass intentionally creates a physical "pendulum effect", significantly enhancing the UAV's passive aerodynamic stability and robustness against wind disturbances during reinforcement learning training.

Table 8: Maximum State Deltas (Δ_{max}) per Communication Interval (0.1s)

State Variable Group	Array Index	Max Kinematic Allowance	Unit
3D Position (x, y, z)	0, 1, 2	± 0.5	m
Euler Angles (ϕ, θ, ψ)	3, 4, 5	± 0.5	rad
Linear Velocity (v_x, v_y, v_z)	6, 7, 8	± 1.0	m/s
Angular Velocity (p, q, r)	9, 10, 11	± 2.0	rad/s

B Appendix B: Comprehensive DRL Hyperparameters and Network Architectures

To ensure the complete reproducibility of the deep reinforcement learning (DRL) experiments presented in this thesis, this appendix details the specific neural network architectures and hyperparameter configurations utilized for the Proximal Policy Optimization (PPO) agent across all validation scenarios.

B.1 Policy and Value Network Architecture

Given the complex, expanded 34-dimensional spatial-aware observation space (incorporating generalized coordinates, velocities, and 3D Euclidean tracking errors), a high-capacity neural architecture was essential to capture the non-linear mapping between spatial tracking and bandwidth distribution. Both the Actor (Policy) network and the Critic (Value) network share a symmetric Multilayer Perceptron (MLP) design, though their weights are completely independent:

- **Input Layer**: Matches the observation dimension (34 for the 7-DOF manipulator, 36 for the 12-dimensional UAV, and 21 for the edge manipulator).
- **Hidden Layers**: Two dense linear layers, each comprising exactly **256 neurons**, intertwined with Rectified Linear Unit (ReLU) activation functions to introduce necessary nonlinearities.
- **Output Layer**: *Actor*: Maps to the action dimension (e.g., 7 for Franka), outputting unbounded continuous values which are subsequently squashed by a deterministic Softmax layer into valid bit distributions; *Critic*: A single neuron outputting the scalar baseline value estimation $V(s)$.

B.2 PPO Hyperparameter Configuration

The training pipeline was executed utilizing the `stable-baselines3` library. To mitigate policy instability during the exploration of highly discontinuous quantization terrains and to suppress physics-induced gradient spikes, the hyperparameters were fine-tuned as listed in Table 9.

Table 9: PPO Algorithm Training Hyperparameters

Hyperparameter	Symbol	Value	Description
Learning Rate	α	3×10^{-4}	Learning rate for the policy and value function optimizers.
Discount Factor	γ	0.99	High emphasis on long-term reward maximization.
GAE Parameter	λ_G	0.95	Generalized Advantage Estimation trace decay parameter for bias-variance trade-off.
Clip Range	ε	0.2	Constrains the surrogate objective ratio to prevent destructively large policy updates.
Entropy Coefficient	c_2	0.01	Encourages sufficient exploration across the discrete quantization step-spaces.
Value Function Coeff.	c_1	0.5	Scaling factor for the Critic's mean squared error loss.
Max Gradient Norm	-	0.5	Global gradient clipping threshold to prevent exploding gradients during kinematic singularities.
Rollout Buffer Size	N_{steps}	512	Steps collected per environment before updating.
Number of Environments	N_{envs}	8 (Multi-core)	Number of parallel PyBullet physics engines executed via `SubprocVecEnv`.
Effective Batch Size	B	4096	Total experiences per update ($N_{steps} \times N_{envs}$).
Reward Clipping Bounds	C	$[-50.0, 50.0]$	Clips the physical mm-level tracking reward to prevent the Critic from experiencing unbounded variance during collisions.

C Appendix C: Digital Twin Configuration of the Edge Manipulator

To validate the proposed Joint Communication and Control (JCC) framework's deployability on resource-constrained edge devices, a high-fidelity Digital Twin of a custom 5-degree-of-freedom (5-DOF) serial manipulator (based on the Jibot1 hardware architecture) was constructed using the Unified Robot Description Format (URDF). This appendix details the precise kinematic dimensions, empirical mass distributions, and the critical soft-servo emulation parameters utilized during the composite stress tests in Section 3.4.

C.1 Kinematic Dimensions and Mass Distribution

Unlike industrial-grade manipulators, edge actuators are characterized by lightweight, cost-effective materials. The total system mass was calibrated to strictly match the physical hardware ($m_{total} = 1.24$ kg). The length of each segment and its corresponding mass are detailed in Table. 10.

Table 10: Kinematic and Inertial Properties of the Custom 5-DOF Manipulator

Link Name	Segment Length (mm)	Calibrated Mass (kg)	Description / Function
`base_link`	60	0.44	Static base platform and yaw motor housing
`link1_bracket`	40	0.20	U-bracket connecting yaw and shoulder pitch
`link2_main_arm`	105 (L_1)	0.25	Primary load-bearing shoulder segment
`link3_forearm`	75 (L_2)	0.15	Secondary elbow segment
`link4_wrist`	65 (L_3)	0.08	Distal wrist pitch articulation
`gripper_assembly`	120 (L_g)	0.12	Combined mass of rotating base and fingers

C.2 Soft-Servo Dynamics and Hardware Compliance Emulation

In theoretical simulations, actuators are frequently modeled with infinite stiffness and instantaneous torque delivery. However, low-cost Internet of Things (IoT) manipulators are typically driven by commercial Pulse-Width Modulation (PWM) servos (e.g., standard 25 kgůcm servos), which exhibit significant mechanical compliance, gear backlash, and torque limitations.

To bridge the Sim-to-Real gap and accurately simulate the "physical sagging" observed when a heavy payload is applied, the physics engine was injected with specific **Soft-Servo Emulation Parameters** during the execution of control commands:

- **Maximum Output Torque (F_{max})**: The maximum output torque for all joints was software-capped at 2.5 $\mathbf{N \cdot m}$ (approximately equivalent to a 25.5 kg·cm servo). This deliberate bottleneck ensures that the unmodeled 0.8 kg payload in the stress test severely overloads the shoulder actuator, forcing the allocation algorithms to prove their adaptive limits.
- **Position Gain (K_p)**: Set to a low value of 0.05. This proportional gain introduces realistic physical compliance into the digital twin. Instead of rigidly locking into position, the simulated joints exhibit a spring-like flexibility, accurately replicating the hardware jitter and trajectory deflection typical of overloaded, low-cost edge robots.

D Appendix D: Pseudo-Code of the Micro-Interpolation Safety Layer

To safely deploy the extremely quantized, low-frequency (10 Hz) control signals generated by the DRL allocator onto physical edge actuators without triggering hardware-damaging torque-derivative faults, a **Micro-Interpolation Safety Layer** is implemented at the edge receiver (e.g., a local robotic microcontroller or PLC).

Algorithm 1 details the core logic of this safety layer. It acts as a local Zero-Order Hold (ZOH) combined with a high-frequency linear interpolator. The layer operates at a strict, synchronous control loop frequency ($f_{ctrl} = 240$ Hz or higher) while receiving sparse network packets asynchronously. This decoupling completely isolates the low-frequency network starvation from the high-frequency mechanical execution.

Algorithm 1 Edge-Node Micro-Interpolation Safety Layer

Input: Asynchronous Network Packets containing quantized target states q_{target}
Input: Current physical joint states q_{current} from hardware encoders
Params: $f_{\text{comm}} = 10$ Hz (Communication rate), $f_{\text{ctrl}} = 240$ Hz (Control rate)
Init: $N_{\text{steps}} = f_{\text{ctrl}}/f_{\text{comm}}$ (Interpolation segments per packet)

while System is Active **do**
    // Asynchronous Receive Callback Routine
    if New_Packet_Received() **then**
        $q_{\text{target}} = \text{Decode\_Payload(Packet)}$
        for joint $i = 1$ to n **do**
            $\Delta q[i] = q_{\text{target}}[i] - q_{\text{current}}[i]$
            $\text{step\_increment}[i] = \Delta q[i]/N_{\text{steps}}$
        end for
    end if

    // High-Frequency Synchronous Control Loop (Runs exactly at f_{ctrl})
    if Hardware_Timer_Tick$(1/f_{\text{ctrl}})$ **then**
        for joint $i = 1$ to n **do**
            $q_{\text{cmd}}[i] = q_{\text{cmd}}[i] + \text{step\_increment}[i]$
            $\tau[i] = \text{Low\_Level\_PID\_Controller}(q_{\text{cmd}}[i], q_{\text{current}}[i])$
            Apply_Actuator_Torque$(i, \tau[i])$
        end for
    end if
end while